1.8 Application on Cardinality of Sets:

We have learnt about the union, intersection, complement and difference of sets.

Now we will go through some practical problems on sets related to everyday life.

Results :

If A and B are two finite sets, then

(i) $n(A \cup B) = n(A)+n(B)- n(A \cap B)$

(ii) $n(A-B) = n(A) - n(A \cap B)$

(iii) $n(B-A) = n(B)- n(A \cap B)$

(iv) $n(A') = n(U) - n(A)$

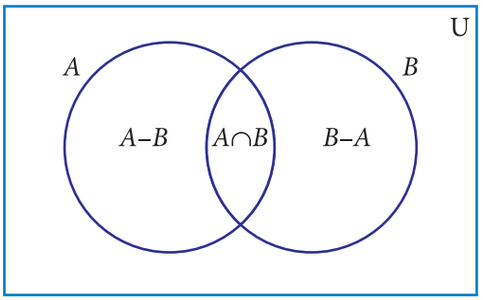


Fig. 1.32

Note

From the above results we may get,

- $n(A \cap B) = n(A)+n(B) - n(A \cup B)$
- $n(U) = n(A)+n(A')$
- If A and B are disjoint sets then, $n(A \cup B) = n(A)+n(B)$.

Example 1.27 From the Venn diagram, verify that $n(A \cup B) = n(A)+n(B) - n(A \cap B)$

Solution From the venn diagram,

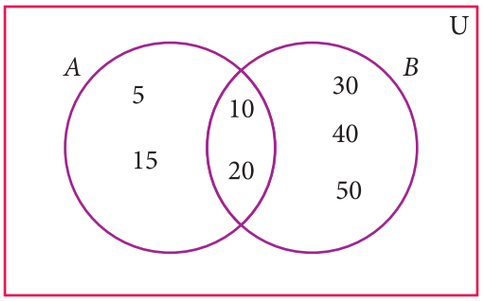


Fig. 1.33

$A = \{5, 10, 15, 20\}$

$B = \{10, 20, 30, 40, 50,\}$

Then $A \cup B = \{5, 10, 15, 20, 30, 40, 50\}$

$A \cap B = \{10, 20\}$

$n(A) = 4$, $n(B) = 5$, $n(A \cup B) = 7$, $n(A \cap B) = 2$

$n(A \cup B) = 7 \quad \rightarrow (1)$

$n(A)+n(B)-n(A \cap B) = 4+5-2$

$=7 \quad \rightarrow (2)$

From (1) and (2), $n(A \cup B) = n(A)+n(B)-n(A \cap B)$.

Example 1.28 If $n(A) = 36$, $n(B) = 10$, $n(A \cup B)=40$, and $n(A')=27$ find $n(U)$ and $n(A \cap B)$.

Solution $n(A) = 36$, $n(B) =10$, $n(A \cup B)=40$, $n(A')=27$

(i) $n(U) = n(A)+n(A') = 36+27 = 63$

(ii) $n(A \cap B) = n(A)+n(B)-n(A \cup B) = 36+10-40 = 46-40 = 6$

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